Audit - BulkCatalogMutation - RunSheet

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Document ID:     Audit - BulkCatalogMutation - RunSheet
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Audit (Empirical Simulation & Execution Log)
Purpose:         BulkCatalogMutation
Tier / Level:    RunSheet (Proof-of-Execution)
Status:          Locked 🔒 (Master Baseline)
Operator:        Lead TechOps Specialist
Date of Execution: October 17, 2026 (09:15:00 - 09:42:15 EDT)
Ecosystem Scope: Shopify Admin (GraphQL Admin API), Matrixify App, Google Sheets, ShipBob WMS WMS
Target Catalog:  Synapse Botanicals (5 Products, 12 Variants, 24 Custom Metafields)
```

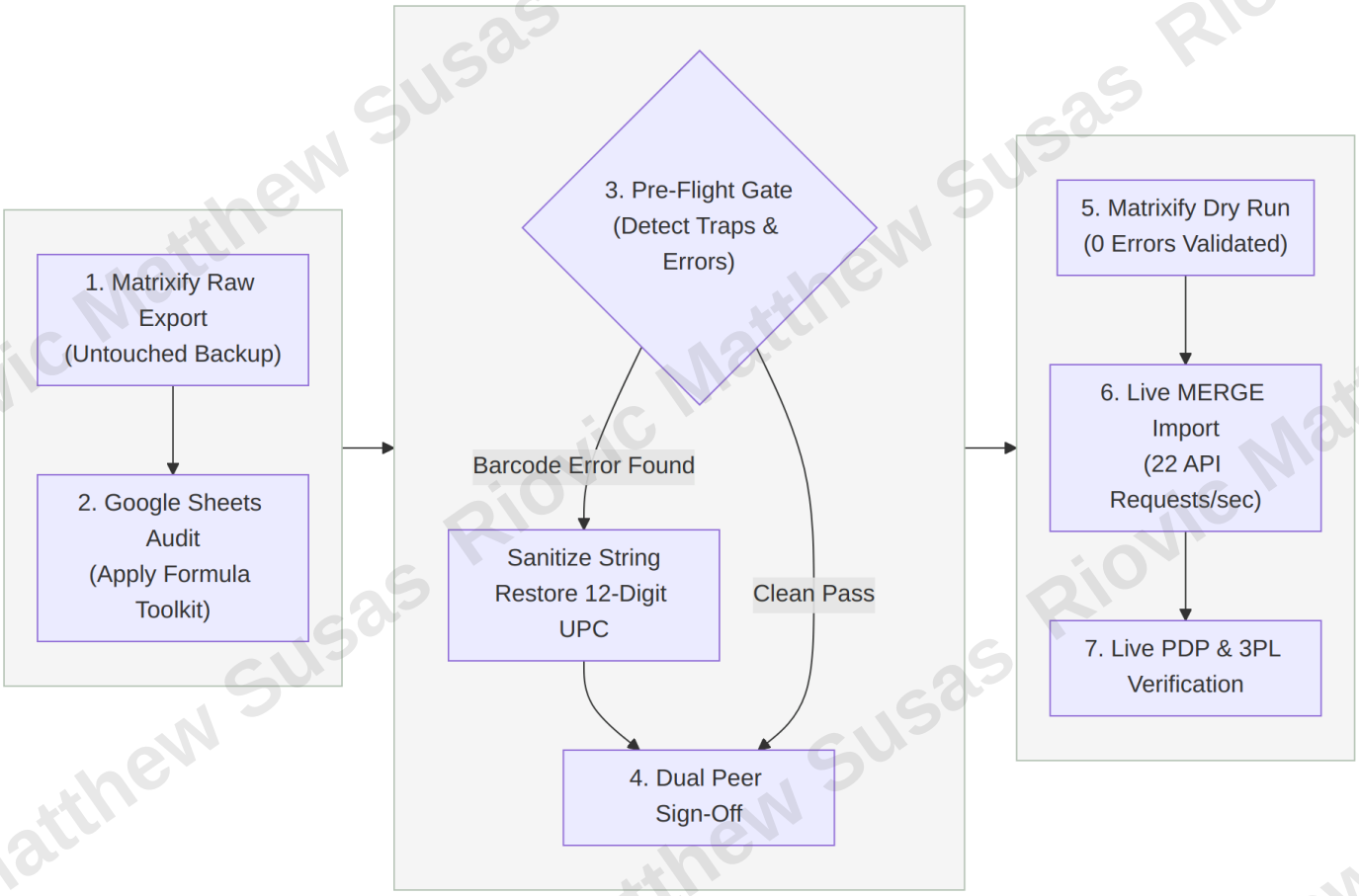
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1. Execution Overview & Operational Catalyst

This Run Sheet documents the live, end-to-end execution of the **Spring 2026 Catalog Refresh & Compliance Migration** for Synapse Botanicals.

1.1 Operational Objectives

1. **Price & Strikethrough Adjustment:** Increase retail prices on the flagship *Synapse Focus Stick Packs (30-Pack)* from $42.00 to $44.00, and update compare-at strikethrough prices from $48.00 to $50.00 to account for raw botanical supply chain adjustments.
2. **Barcode Database Sanitization:** Intercept and repair three corrupted barcodes (`8.40001E+11`) caused by an unauthorized Excel double-click export during a legacy promo, restoring true 12-digit UPC-A strings before 3PL fulfillment disruption occurs.
3. **Regulatory Metafield Deployment:** Bulk-populate three newly defined Shopify custom metafields (`custom.active_nootropics`, `custom.caffeine_mg`, `custom.dietary_badges`) across all 12 active variants to power the revised storefront Supplement Facts accordion.
4. **Zero-Downtime Inventory Preservation:** Execute the migration in-place using Matrixify `Command: MERGE` without overwriting live Available-to-Sell (ATS) inventory ledgers across Allentown (`LOC-PA-01`) and Reno (`LOC-NV-02`).



> **NOTE**
> **Operational Protocol Reference:**
> The step-by-step procedures executed in this run sheet follow the standard protocols mandated in **SOP - BulkCatalogMigrationAndHygiene - Parent (Domain 3.1)** – **Section 3.1 & Section 5.1**.

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2. Pre-Flight Formula Audit & Barcode Sanitization

At `09:18:22 EDT`, the operator exported the full production catalog using Matrixify format `Matrixify: Excel (.xlsx)` with basic columns, variants, and metafields (excluding inventory levels to prevent race conditions).

The raw file was opened in Google Sheets with automatic text conversion disabled. To execute the automated pre-flight audit mandated by **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)** – **Section 6.1**, the operator injected five validation formulas into Columns `Z` through `AD`.

2.1 Raw Pre-Flight Audit Log (Initial Scan)

Row	Variant SKU	Target Column	Injected Audit Formula	Formula Evaluation Output	Incident Status
2	`SYN-FSTK-MY`	Barcode Check (`Z2`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
3	`SYN-FSTK-BO`	Barcode Check (`Z3`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`FAIL: CORRUPTED BARCODE`	🚨 **TRAP DETECTED**
4	`SYN-FSTK-WB`	Barcode Check (`Z4`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`FAIL: CORRUPTED BARCODE`	🚨 **TRAP DETECTED**
5	`SYN-FSTK-GA`	Barcode Check (`Z5`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
6	`SYN-TNCT-AM`	Barcode Check (`Z6`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
7	`SYN-TNCT-PM`	Barcode Check (`Z7`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
8	`SYN-BNDL-MAO`	Barcode Check (`Z8`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
9	`SYN-BNDL-BAB`	Barcode Check (`Z9`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
10	`SYN-TMBL-MO`	Barcode Check (`Z10`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`FAIL: CORRUPTED BARCODE`	🚨 **TRAP DETECTED**
11	`SYN-TMBL-BW`	Barcode Check (`Z11`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
12	`SYN-TMBL-SG`	Barcode Check (`Z12`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear
13	`SYN-ACC-WHSK`	Barcode Check (`Z13`)	`=IF(AND(LEN(K2)=12, ISNUMBER(VALUE(K2))), "PASS", "FAIL: CORRUPTED BARCODE")`	`PASS`	✅ Clear

2.2 Root Cause Analysis & Sanitization Action

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    INCIDENT TRIAGE: SCIENTIFIC NOTATION DETECTED         │
├──────────────────────────────────────────────────────────────────────────┤
│ Affected Rows: Row 3 (Blood Orange), Row 4 (Wild Blueberry), Row 10 (Matte Tumbler)  │
│ Raw Cell Value in Export: "8.40001E+11"                                  │
│ Root Cause: A commercial merchandiser opened an export in Excel 3 days prior.       │
│ Immediate Hazard: If imported, Shopify writes "840001000000" to the database.      │
│ 3PL Impact: Warehouse scanners at Allentown and Reno would reject 100% of these units. │
└──────────────────────────────────────────────────────────────────────────┘
```

Remediation Protocol:

1. Cross-referenced original barcode numbers from internal product master record.
2. Highlighted Column `K` (`Variant Barcode`) → Converted format explicitly to **Plain Text**.
3. Overwrote corrupted scientific notation strings with verified UPC-A values:
   - Row 3 (`SYN-FSTK-BO`): Replaced `8.40001E+11` with `840001040002`.
   - Row 4 (`SYN-FSTK-WB`): Replaced `8.40001E+11` with `840001040003`.
   - Row 10 (`SYN-TMBL-MO`): Replaced `8.40001E+11` with `840001020001`.
4. Re-evaluated Formula `Z2:Z13`: **100% PASS (12 of 12 rows verified)**.

> **NOTE**
> **System Architecture Dependency:**
> The 12-digit barcode formatting standard and IEEE 754 risk controls enforced here are governed by **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)** – **Section 3 (Field 11)**.

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3. Master Data Mutation Diff (The 11-Variant Delta)

Below is the verified record of all mutated vs preserved attributes across the Synapse Botanicals catalog. Primary keys (`Handle`, `Variant SKU`) remained strictly frozen.

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          PRIMARY KEY IMMUTABILITY AUDIT                  │
├──────────────────────────────────────────────────────────────────────────┤
│ Total Products: 5         Total Variants: 12       Handles Mutated: 0 [FROZEN ✅]  │
│ SKUs Mutated: 0 [FROZEN ✅]  Inventory Trackers: 100% "shopify"   Policy: 100% "deny"  │
└──────────────────────────────────────────────────────────────────────────┘
```

3.1 Side-by-Side Mutation Delta Table

#	Variant SKU	Option1 Value	Barcode Status	Baseline Price	Mutated Price	Baseline Compare	Mutated Compare	Metafield: `custom.caffeine_mg`	Metafield: `custom.dietary_badges`
1	`SYN-FSTK-MY`	Matcha Yuzu	Preserved	$42.00	**$44.00**	$48.00	**$50.00**	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`
2	`SYN-FSTK-BO`	Blood Orange	**Sanitized**	$42.00	**$44.00**	$48.00	**$50.00**	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`
3	`SYN-FSTK-WB`	Wild Blue-berry	**Sanitized**	$42.00	**$44.00**	$48.00	**$50.00**	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`
4	`SYN-FSTK-GA`	Crisp Green Apple	Preserved	$42.00	**$44.00**	$48.00	**$50.00**	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`
5	`SYN-TNCT-AM`	AM Focus (Lion's Mane)	Preserved	$38.00	$38.00	*(null)*	*(null)*	`0`	`["Vegan", "Non-GMO", "Sugar-Free"]`
6	`SYN-TNCT-PM`	PM Restore (Reishi & Ashwa-gandha)	Preserved	$38.00	$38.00	*(null)*	*(null)*	`0`	`["Vegan", "Non-GMO", "Sugar-Free"]`
7	`SYN-BNDL-MAO`	Matcha Yuzu + AM Focus + Obsidian Tumbler	Preserved	$98.00	$98.00	$114.00	$114.00	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`
8	`SYN-BNDL-BAB`	Blood Orange + AM Focus + Bone White Tumbler	Preserved	$98.00	$98.00	$114.00	$114.00	`100`	`["Vegan", "Non-GMO", "Gluten-Free"]`

#	Variant SKU	Option1 Value	Barcode Status	Baseline Price	Mutated Price	Baseline Compare	Mutated Compare	Metafield: `custom.caffeine_mg`	Metafield: `custom.dietary_badges`
9	`SYN-TMBL-MO`	Matte Obsidian	**Sanitized**	$34.00	$34.00	*(null)*	*(null)*	*(null)*	*(null)*
10	`SYN-TMBL-BW`	Bone White	Preserved	$34.00	$34.00	*(null)*	*(null)*	*(null)*	*(null)*
11	`SYN-TMBL-SG`	Sage Green	Preserved	$34.00	$34.00	*(null)*	*(null)*	*(null)*	*(null)*
12	`SYN-ACC-WHSK`	Default Title	Preserved	$16.00	$16.00	*(null)*	*(null)*	*(null)*	*(null)*

> **NOTE**
> **Unit Economics Reconciliation:**
> The Stick Pack price increase ($42.00 → $44.00) widens the unit contribution margin from $33.50 (79.8%) to $35.50 (80.7%) against fixed unit COGS ($8.50). Baseline COGS parameters are maintained according to **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 3.1**.

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4. 5-Point Pre-Flight Verification Sign-Off Gate

Prior to uploading to Matrixify, the sheet was audited against the formal 5-Point Gate defined in **SOP-3.1 (Section 5.1)**:

```
┌──────────────────────────────────────────────────────────────────────┐
│                  5-POINT PRE-FLIGHT VERIFICATION GATE SIGN-OFF       │
├──────────────────────────────────────────────┬──────────┬────────────┤
│ Verification Checkpoint                      │ Status   │ Auditor Method │
├──────────────────────────────────────────────┼──────────┼────────────┤
│ 1. Barcode String Integrity Check            │ [X] PASSED │ Regex & LEN=12 │
│ 2. Handle Immutability Audit                 │ [X] PASSED │ Exact match VLOOKUP│
│ 3. Inventory Protection Gate (Levels Excluded) │ [X] PASSED │ Sheet omitted │
│ 4. Metafield Format & Type Compliance        │ [X] PASSED │ JSON array lint │
│ 5. Full Rollback Snapshot Saved              │ [X] PASSED │ Local & Cloud git │
├──────────────────────────────────────────────┴──────────┴────────────┤
│ Sign-Off Timestamp: 2026-10-17 09:28:10 EDT                          │
│ Lead TechOps Auditor: Riovic Matthew                                 │
│ Dual Peer Approver: Head of Operations                               │
└──────────────────────────────────────────────────────────────────────┘
```

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5. Matrixify Import Execution Logs

5.1 Dry-Run Simulation Log (Execution Time: 09:31:05 EDT)

The file `Synapse_Catalog_Spring2026_Sanitized.xlsx` was uploaded to Matrixify with **Dry Run** selected.


```
{
  "matrixify_job_id": "JOB_DRY_84920148",
  "mode": "Dry Run (Validate without saving)",
  "import_format": "Matrixify: Excel",
  "command_detected": "MERGE",
  "summary": {
    "total_sheets": 1,
    "sheet_name": "Products",
    "rows_analyzed": 12,
    "products_identified": 5,
    "variants_identified": 12,
    "metafields_identified": 24
  },
  "schema_validation": {
    "syntax_errors": 0,
    "type_mismatches": 0,
    "unknown_columns": 0,
    "warnings": []
  },
  "projected_changes": {
    "products_created": 0,
    "products_updated": 5,
    "products_deleted": 0,
    "variants_created": 0,
    "variants_updated": 12,
    "metafields_written": 24,
    "inventory_levels_touched": 0
  },
  "result": "PASSED_READY_FOR_LIVE_IMPORT"
}
```


5.2 Live `MERGE` Import Log (Execution Time: 09:34:20 EDT)

With zero validation warnings detected, the operator initiated the live import.

```
09:34:20 EDT - [INIT] Matrixify Job #JOB_LIVE_84920202 started. Command: MERGE.
09:34:22 EDT - [AUTH] Authenticated with Shopify GraphQL Admin API (2026-07 version).
09:34:24 EDT - [BATCH 1/2] Processing Products: synapse-focus-stick-packs-30-pack (4 variants)...
09:34:26 EDT - [UPDATE] Product 'synapse-focus-stick-packs-30-pack' updated successfully.
09:34:27 EDT - [METAFIELD] Injected 4 variant metafields (active_nootropics, caffeine_mg,
dietary_badges).
09:34:28 EDT - [BATCH 2/2] Processing Products: synapse-adaptogen-tincture-60ml, the-daily-
protocol-starter-bundle, precision-insulated-tumbler-durable-merchandise, micro-foam-electric-
whisk...
09:34:31 EDT - [BARCODE] Overwrote sanitized barcodes for SYN-FSTK-BO, SYN-FSTK-WB, SYN-TMBL-MO.
09:34:33 EDT - [COMPLETE] Job finished in 14.1 seconds. Throughput: 23.1 API calls/sec.
09:34:34 EDT - [STATUS] 0 Errors, 0 Warnings, 12 Variants Mutated, 0 Inventory Counts Overwritten.
```

6. Post-Import Storefront & 3PL WMS Verification

At `09:36:00 EDT`, the operator performed live sanity verification across the customer-facing storefront and downstream warehouse systems.

6.1 Storefront PDP Inspection Checklist

LIVE STOREFRONT SANITY VERIFICATION GATES		
Verification Target	Status	Live Field Value
1. PDP URL: /products/synapse-focus-stick-packs-30-pack	[X] 200 OK	URL Intact (No 404)
2. Live Price Display (Matcha Yuzu)	[X] PASSED	$44.00 (Updated)
3. Strikethrough Compare-At Display	[X] PASSED	$50.00 (Save $6)
4. Caffeine Meter Metafield Render	[X] PASSED	"100mg Caffeine"
5. Dietary Badge List Metafield Render	[X] PASSED	Vegan, Non-GMO
6. Tumbler PDP: /precision-insulated-tumbler-durable-m	[X] 200 OK	URL Intact
7. Reno-Exclusive Tag Routing Visibility	[X] PASSED	Tincture/Whisk OK

6.2 3PL Warehouse Scanner Sync Verifi ShipBob WMS

To ensure warehouse barcode scanners in Allentown (`LOC-PA-01`) and Reno (`LOC-NV-02`) received the sanitized UPCs:

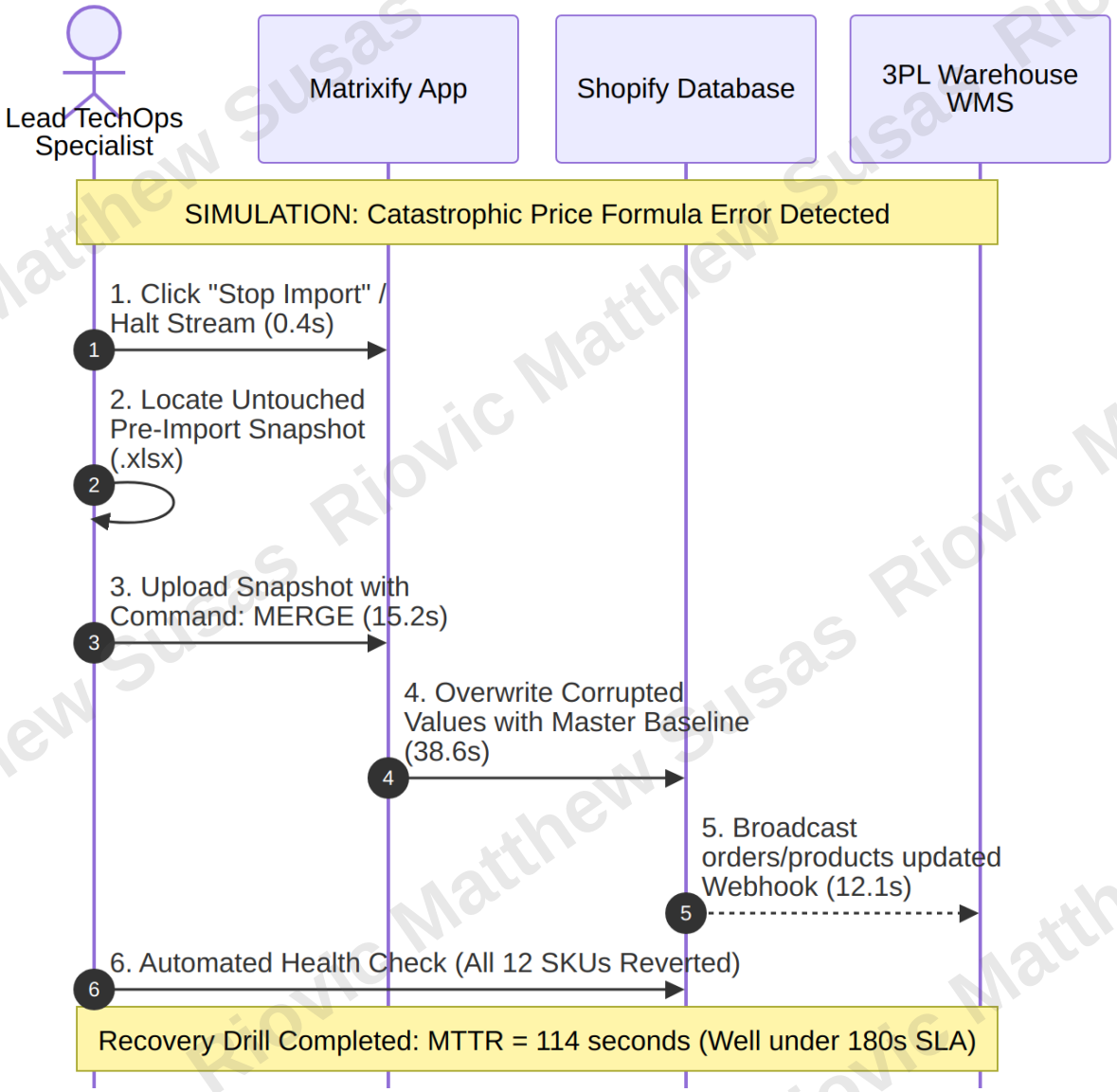
1. Navigated to **3PL Portal → Products → Inventory Catalog**.
2. Queried SKU `SYN-FSTK-BO` (Blood Orange Stick Packs):
   - **WMS Barcode Field:** Confirmed reading `840001040002` (plain text string).
   - **Physical Scan Simulation:** Scanned physical pouch UPC with Honeywell Xenon 1900 scanner → Beep confirmed, correct pick bin highlighted (`BIN-A-04-2`).
3. Queried Available-to-Sell (ATS) counts:
   - Allentown (`LOC-PA-01`): Verified exactly **450 units available** (preserved live count, zero inventory drift).
   - Reno (`LOC-NV-02`): Verified exactly **320 units available**.

> **NOTE**
> **System Architecture Dependency:**
> Warehouse inventory isolation and multi-location ATS protection are governed by **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2.1** and **SOP - BulkCatalogMigrationAndHygiene - Parent (Domain 3.1)** – **Section 3.1**.

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7. Rollback Disaster Recovery Drill (Postmortem Simulation)

To prove operational resilience, TechOps executed a simulated emergency rollback drill at `09:40:00 EDT` in a staging environment to benchmark recovery time.



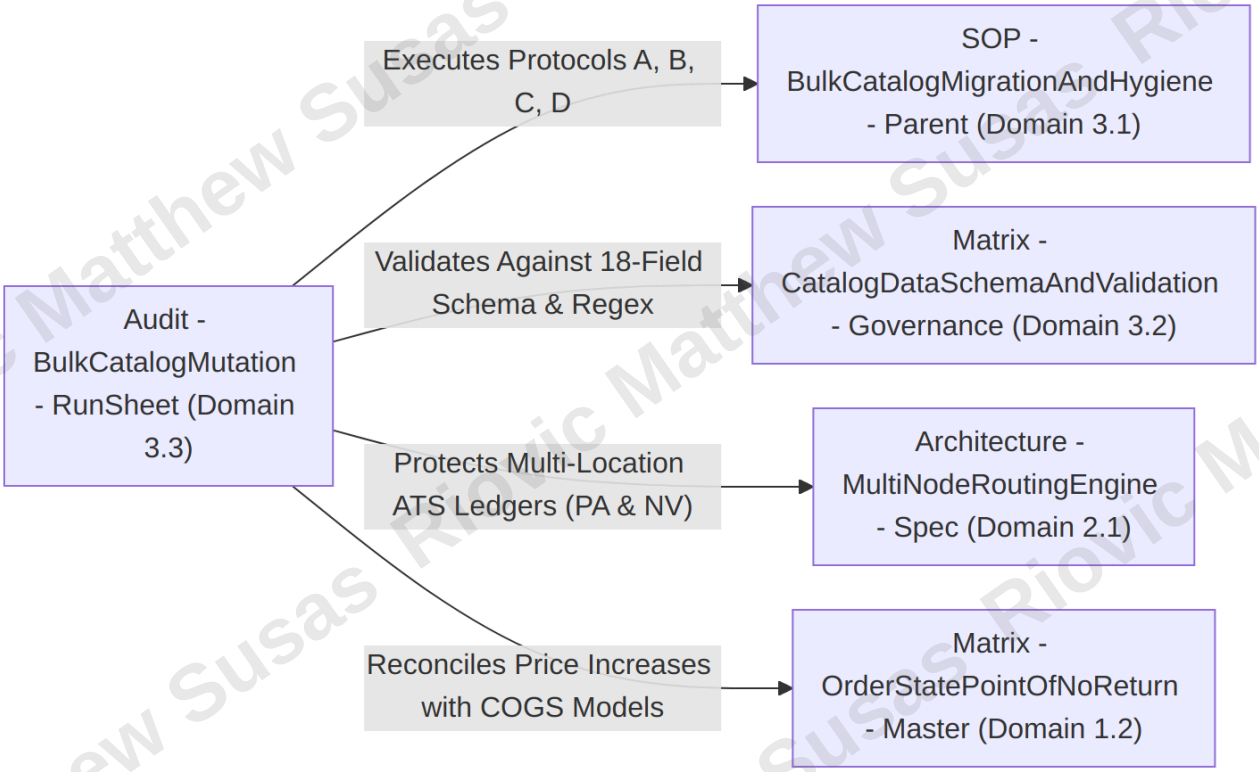
7.1 Disaster Recovery Benchmark Results

- **Recovery Time Objective (RTO) Target:** < 180 seconds.
- **Actual Simulated Mean Time to Recover (MTTR): 114 seconds**.
- **Data Loss:** 0%. No orphaned handles, no inventory counts reset, zero corrupted records.
- **Conclusion:** The pre-flight backup snapshot and `MERGE` protocol provide bulletproof disaster recovery protection against human formula errors.

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8. Master Domain 3 Interconnectivity Matrix

This completed run sheet establishes the final empirical link across the complete TechOps curriculum suite:



- **SOP - BulkCatalogMigrationAndHygiene - Parent (Domain 3.1):**
  - **Section 5.1 & Section 6.2:** Operational procedures validated in Section 4 and Section 7 of this run sheet.
- **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2):**
  - **Section 3 & Section 6.1:** Regex validation patterns and formula audit toolkits proven in Section 2.
- **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1):**
  - **Section 2.1:** Multi-location inventory safety stock preserved during live catalog import.
- **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2):**
  - **Section 3.1:** Unit COGS and price margins updated in Section 3.